PUBLIC CICR SUSTAINABILITY ASSESSMENT TOOL MANUAL

(LAST UPDATED: 3/5/2026)

CÁMARA DE INDUSTRIAS DE COSTA RICA

MÁS COMPETITIVIDAD, MÁS EMPLEO

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HOW TO USE THE PRODUCT SUSTAINABILITY ASSESSMENT TOOL

How to Fill One Out

General Overview:

This tool has been crafted to provide guidance for the improvement of sustainability for products and their manufacturing practices. It involves a questionnaire portion which will require input regarding a singular product of one's own choice. Once completed, it will output a chart containing scores in 6 categories of sustainability. A progress bar at the top shows the question and category you are on.

This tool is comprised of approximately 44 total questions (as of March 2026) which will take 20-30 minutes to finish. The tool is broken up into 4 main sections:

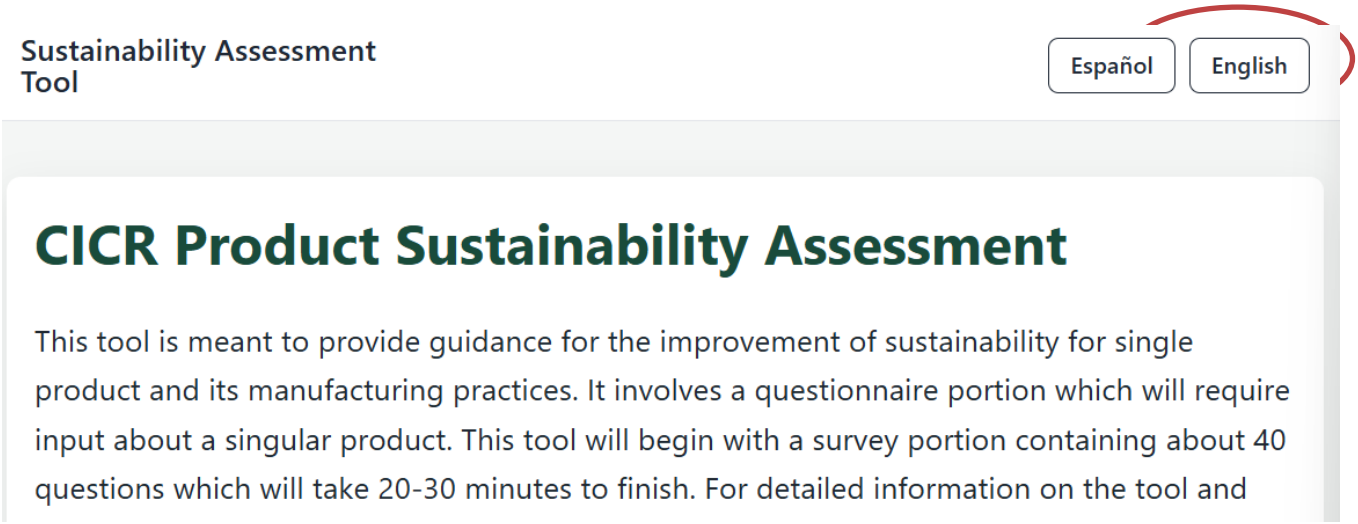
1. Landing Page – This is the starting page of the tool. It consists of a short explanation of what the tool is, how it works, and a confidentiality statement. Moving past this stage acts as a electronic signature.
2. Demographic Questions – The first few questions will be demographic focused and are for you and CICR's record keeping.
3. Scored Questions – Questions can only increase your score, and no answer will subtract from your total score. Each question will have the option to answer, "Not applicable/Not sure", and that question will not be used in the final score calculation.
4. Results Page – After submission, a radar chart will be displayed, along with the scores of your product for each of the categories. You will have the option to download a copy of your output for your internal use.

Reloading or closing the tool pass the landing page will not save your answers and you will be brought back to the landing page.

Confidentiality:

Information that is collected or created by this tool will be saved to a CICR database and will only be accessible to select CICR representatives. CICR will use this data for statistical reporting and will not publish any identifying information regarding your product, processes, or company. All information will be treated with the strictest confidentiality, and the results will be presented only in an aggregate manner. The option to share your results beyond this remains solely with your company. Results will not be saved until the final question is answered. Continuing past this page acts as your electronic consent signature for these terms.

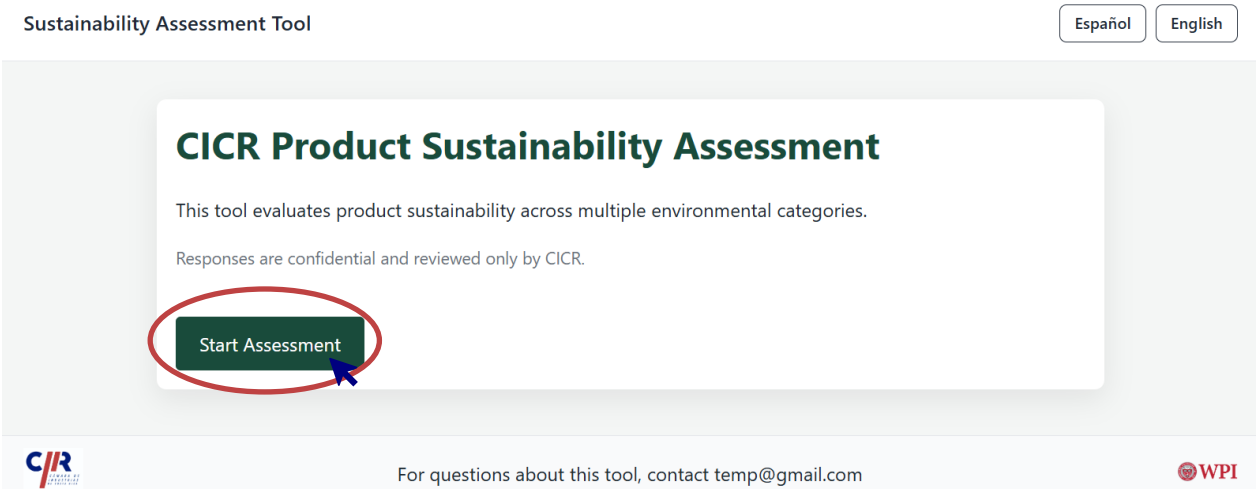
Switching Languages:



The language of this tool can easily be swapped between Spanish and English at any point. The language will live update, so any previous responsive will also be swapped to desired language.

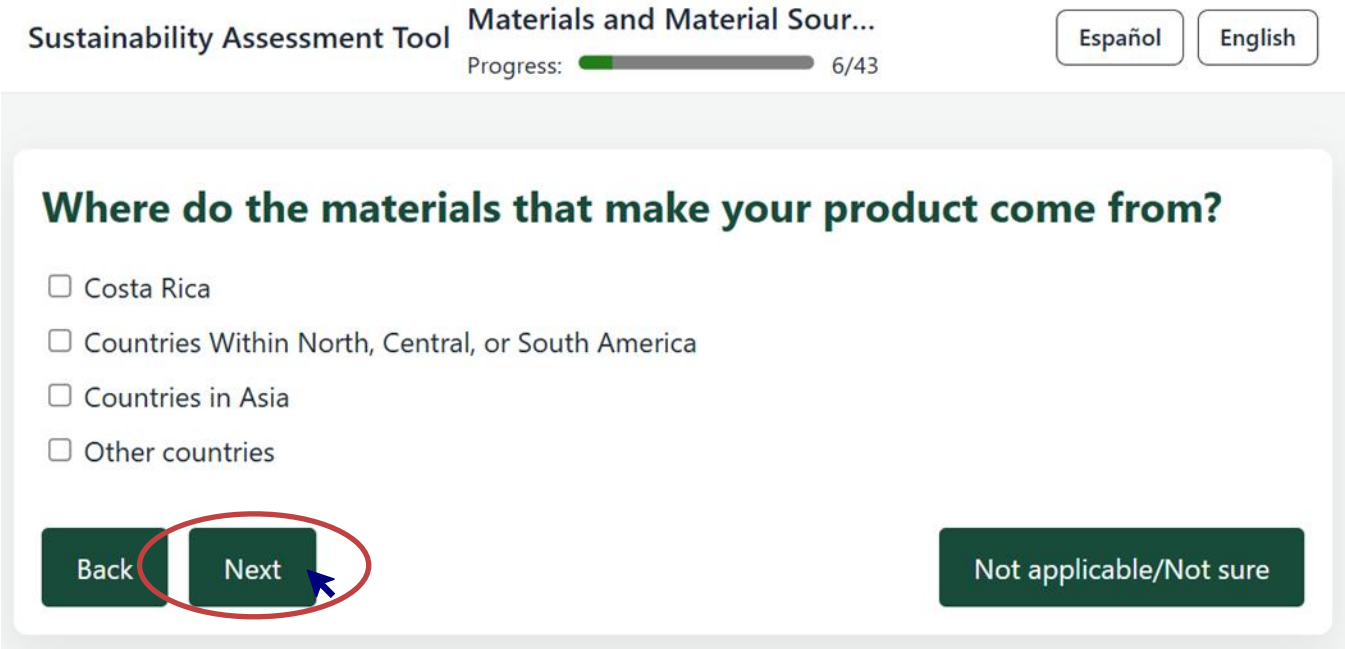
Question Types and How to Answer Them:

How to Start:



To start the assessment, click the button "Start Assessment." Once this button is clicked, the assessment has started. If you close or reload the tool, answer will not be saved, and you will be brought back to this page.

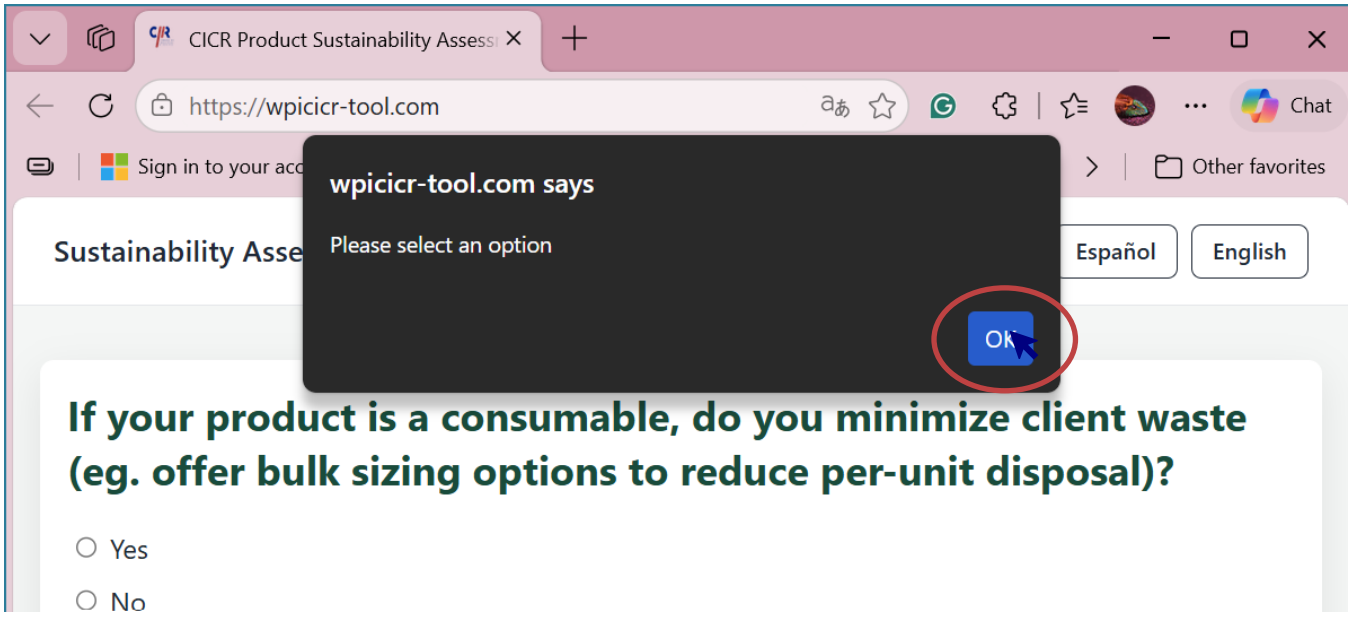
How to Submit an Answer/Finish the Tool:



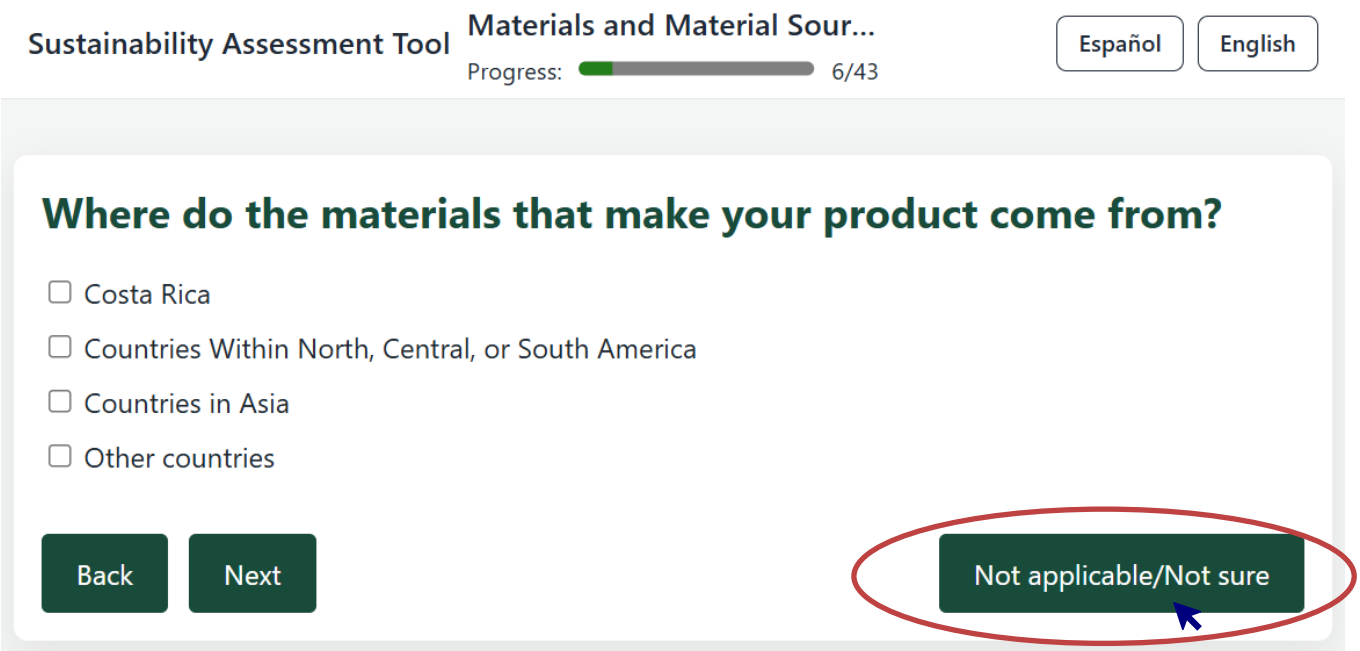
To submit any answer hit the "Next." If no answer is selected, an error message will pop up. Close the error message and input a response. To submit the entire tool, just hit the final "Next" button on

question 44. You can see your progress bar and the category you are in at the top of the page. That will automatically bring you to the results page and submit the answers to CICR.

Error Message:

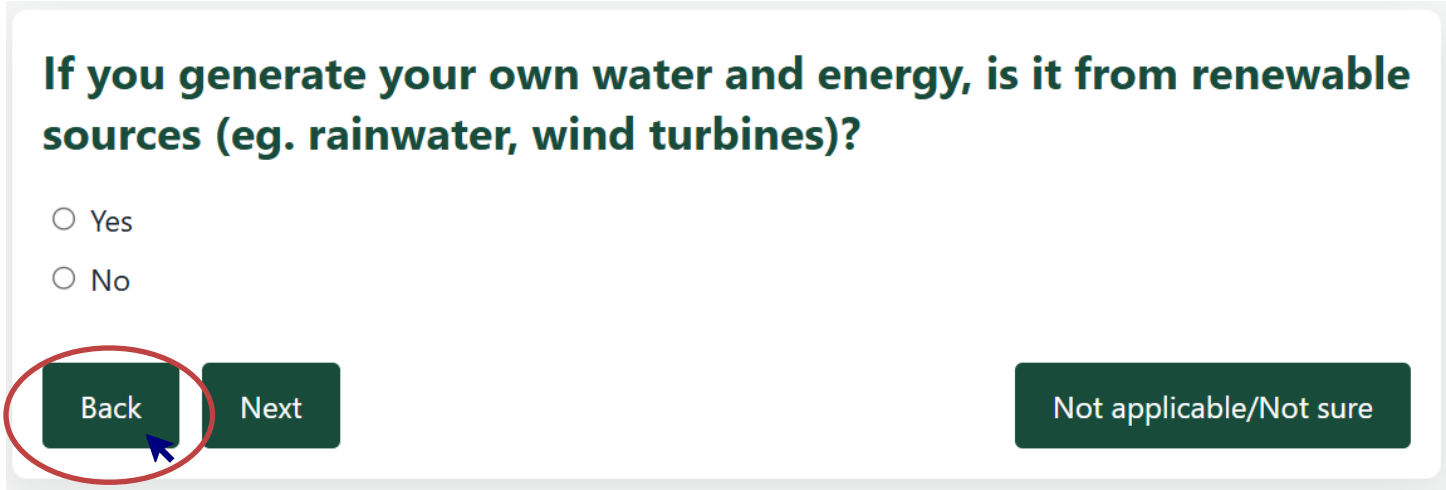


How to Skip a Question:



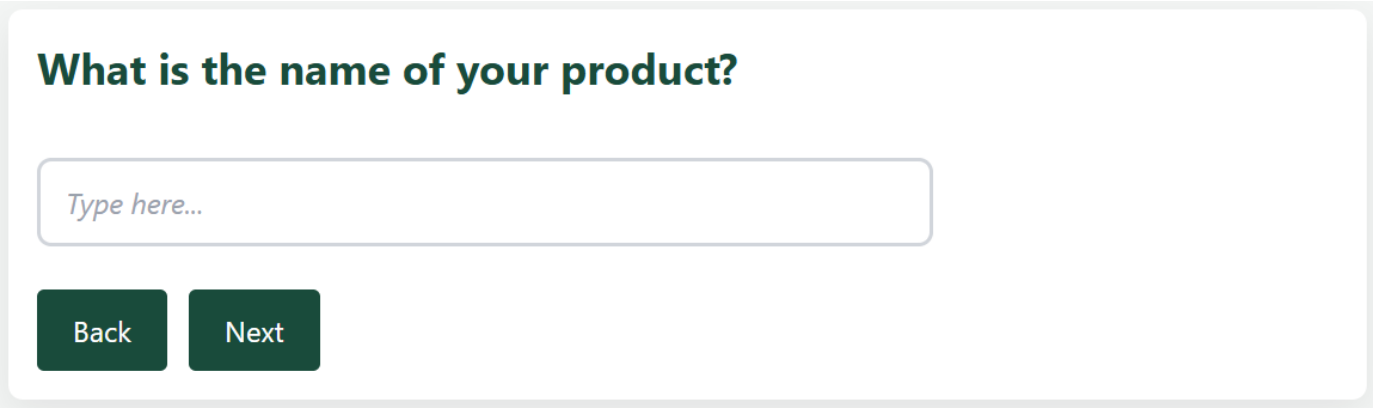
If a question does not feel applicable or one is not sure of the answer, the question can be skipped and not include in final score calculations through the “Not applicable/Not sure” button.

How to go Back:



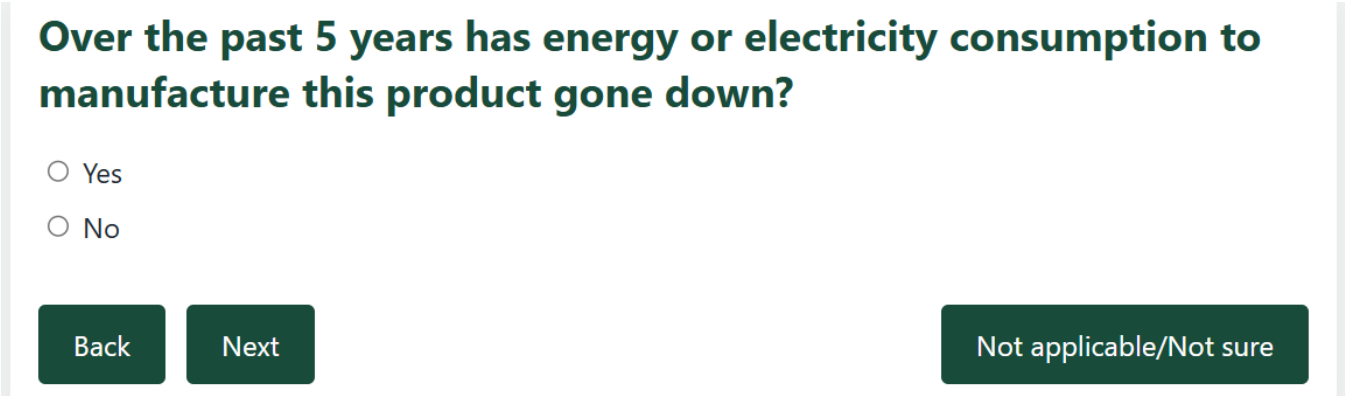
To go back, just hit the “Back” button. Your previous answer will be shown, and you can change your response.

Textbox:



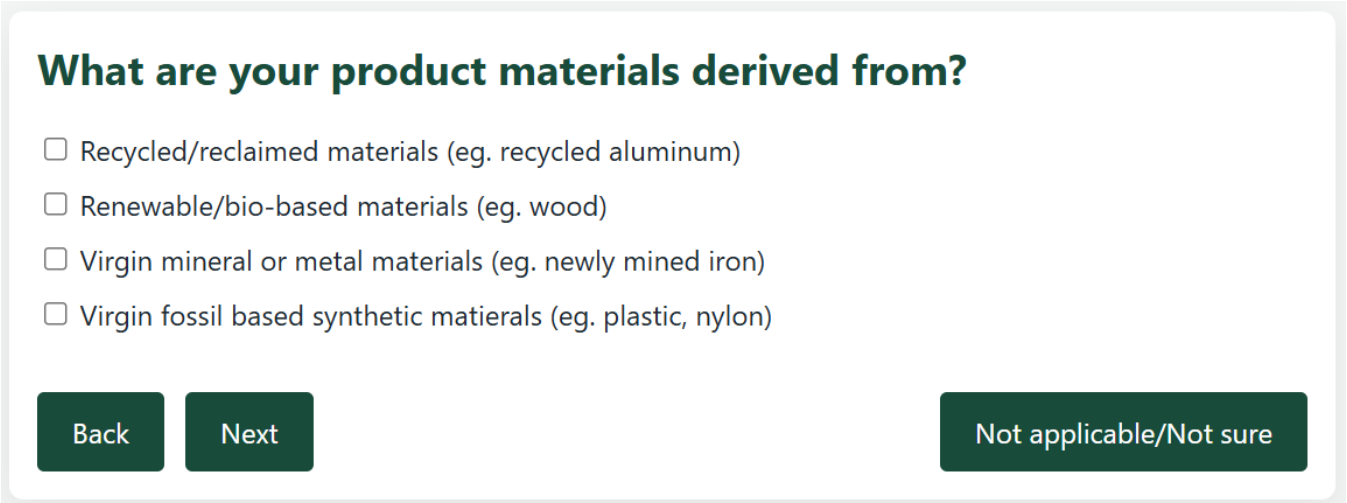
To answer, click on the text box and begin typing. To submit the answer, hit next. These questions will save whatever is typed as the answer. These answers will not be used to calculate scores.

Multiple-choice:



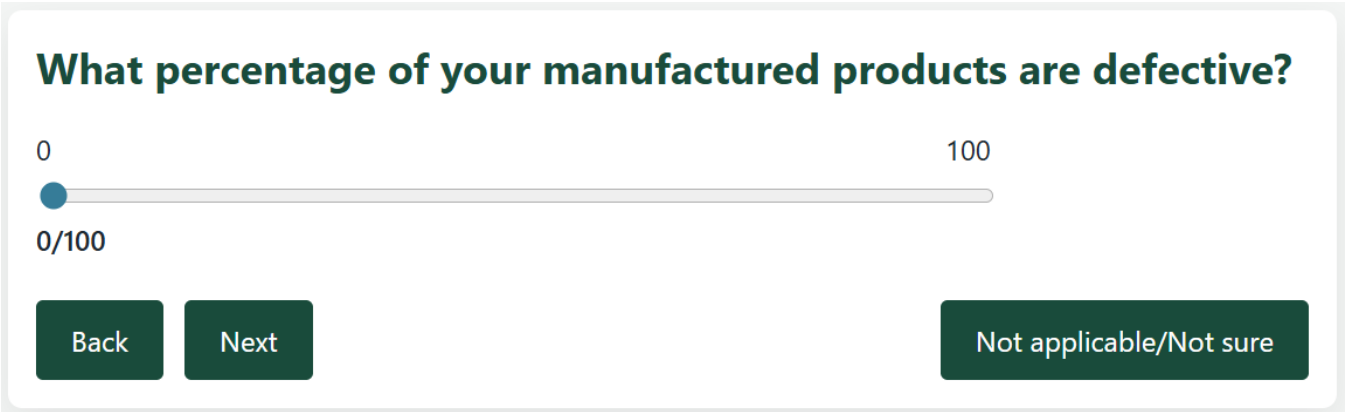
To answer, select one of the answers by either selecting the corresponding circle or the words of the answers itself. To submit the answer, hit next. The option you selected will be saved as your answer. Besides the demographic questions, these questions are used to calculate the product's scores with the score corresponding to your answer selection.

Checkbox:



To answer, select one or more of the answers by either selecting the corresponding square or the words of the answers itself. To submit the answer, hit next. The options you selected will be saved as your answer. Besides the demographic questions, these questions are used to calculate the product's scores. One total score for a question corresponds to your answer selections individual scores added together.

Slider:

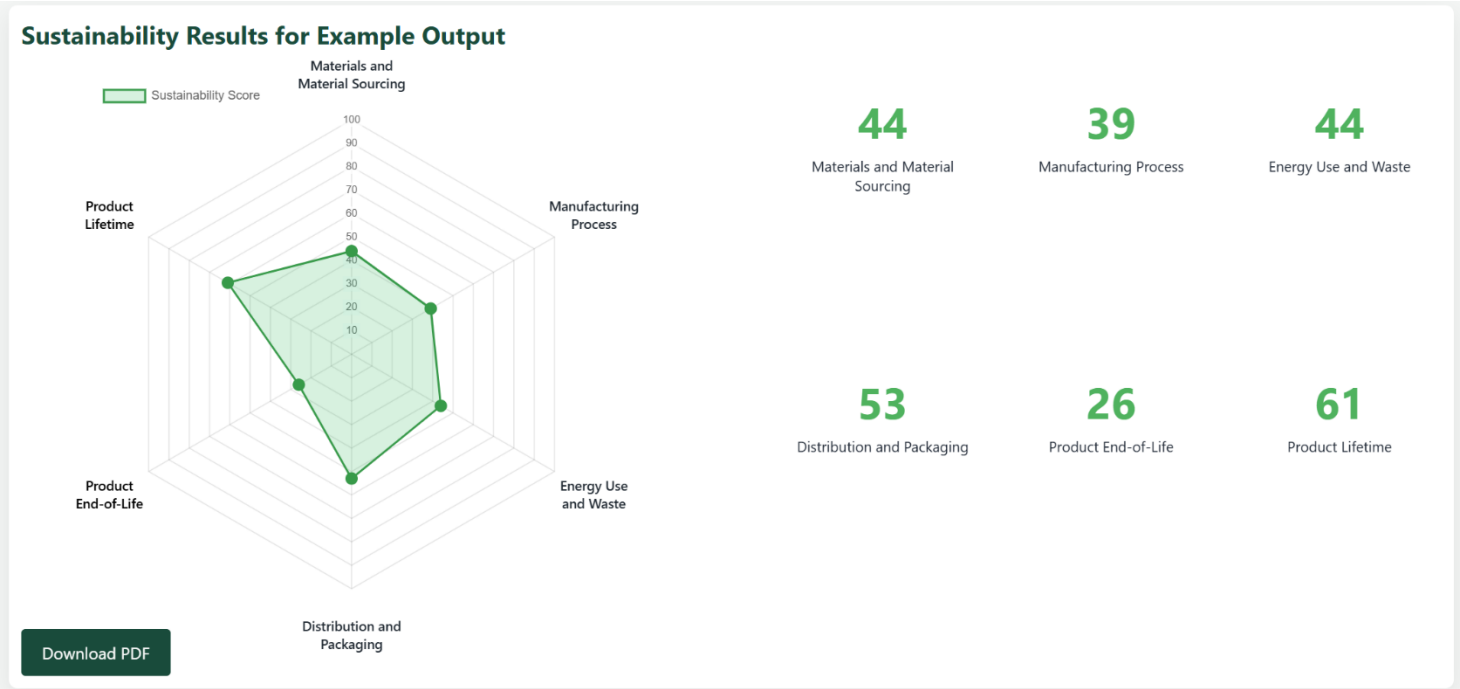


To answer, either select the blue circle and drag, or click anywhere along the slider line. Each slider will have a minimum and maximum corresponding to either end of the slider. To submit your answer, hit next. Besides the demographic questions, these questions are used to calculate the product's scores. The

score is proportional to the percentage of the distance from the minimum to the maximum that the slider is set at.

How to Interpret the Output

Example Output:



Above, you can see an example of the final output generated after completing the assessment. The page consists of three main components: a dynamic title, a radar chart, and a score panel. The title automatically includes the product name entered at the beginning of the assessment, allowing each report to be clearly identified and associated with a specific product.

The radar chart visually represents performance across the six evaluation categories: Materials and Material Sourcing, Manufacturing Process, Energy Use and Waste, Distribution and Packaging, Product Lifetime, and Product End-of-Life. Each axis corresponds to one category and displays the normalized score (0–100). The plotted shape provides a quick visual summary of strengths and areas for improvement across the product's lifecycle.

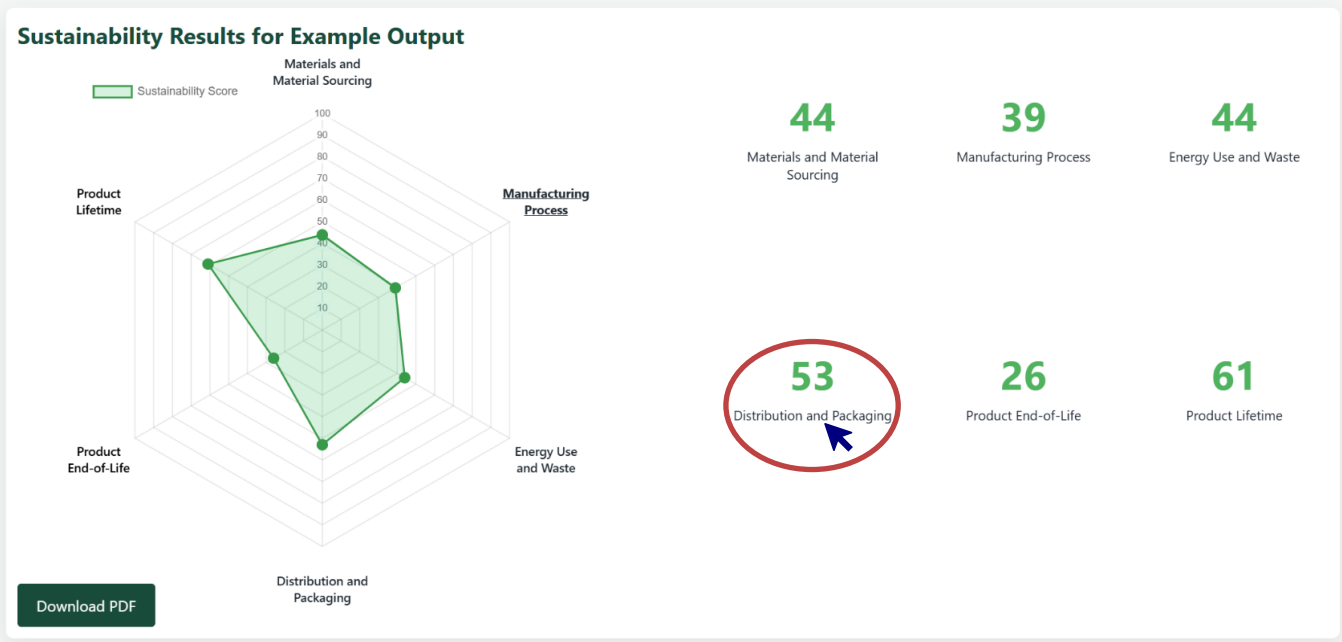
Next to the chart, the score panel displays the exact numerical score for each category. These values allow for precise comparison and documentation beyond the visual representation shown in the chart.

Once you have reached this page, your responses have been successfully submitted to CICR. The results displayed reflect the information provided throughout the assessment and serve as a summary of your product's sustainability performance.

Displaying Category Information:

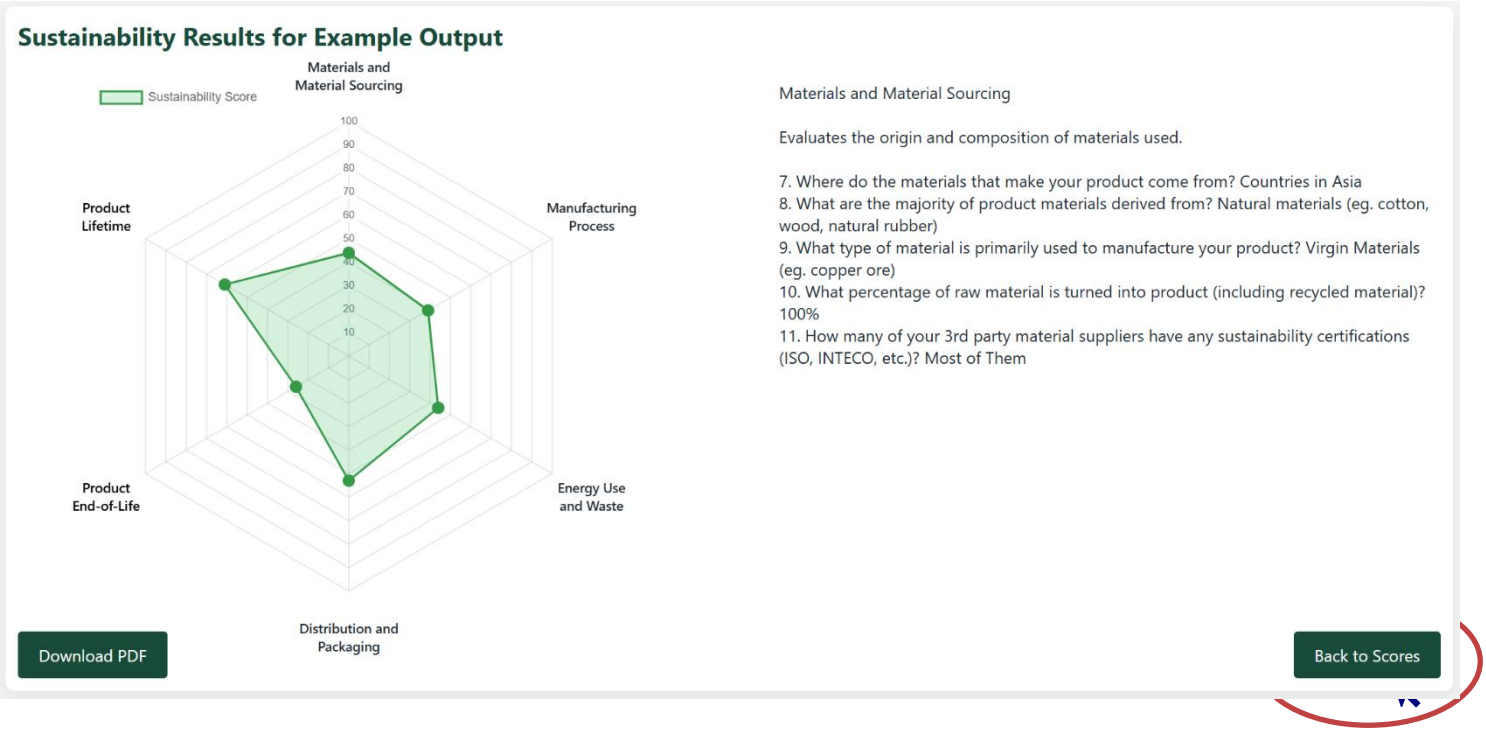
Additional information about each category, including which specific questions contribute to that category's score, can be accessed in two ways. The first method is by clicking directly on a category within the radar chart:

The second way is to click the categories on the score panel itself:



Once a category is selected, a detailed view will appear. This page displays the category title, a short description of the category, the list of questions associated with it, and your selected answer for each question. This breakdown allows you to clearly see how your responses contributed to that category's score.

To return to the main results page and score panel, click “Back to Scores.”



How to Interpret the Radar Chart and Scores:

Each question in the assessment is worth 10 points. Because each category contains a different number of questions, raw point totals are normalized to a 0–100 scale to ensure fair comparison across categories. This means that regardless of how many questions a category contains, its final score is presented as a percentage of the total possible points for that category. For example, if a category contains five questions (50 possible points total) and a product earns 40 points, the final normalized score for that category would be 80 out of 100. This normalization ensures that no category is weighted more heavily simply because it includes more questions. A score of 100 represents the maximum achievable performance in that category, while a score of 0 represents the lowest.

The radar chart visually displays these normalized category scores. Each axis represents one category, and values increase from the center (0) outward to the edge (100). The plotted shape connects the scores across all categories. Points farther from the center indicate stronger performance in that area, while points closer to the center indicate opportunities for improvement.

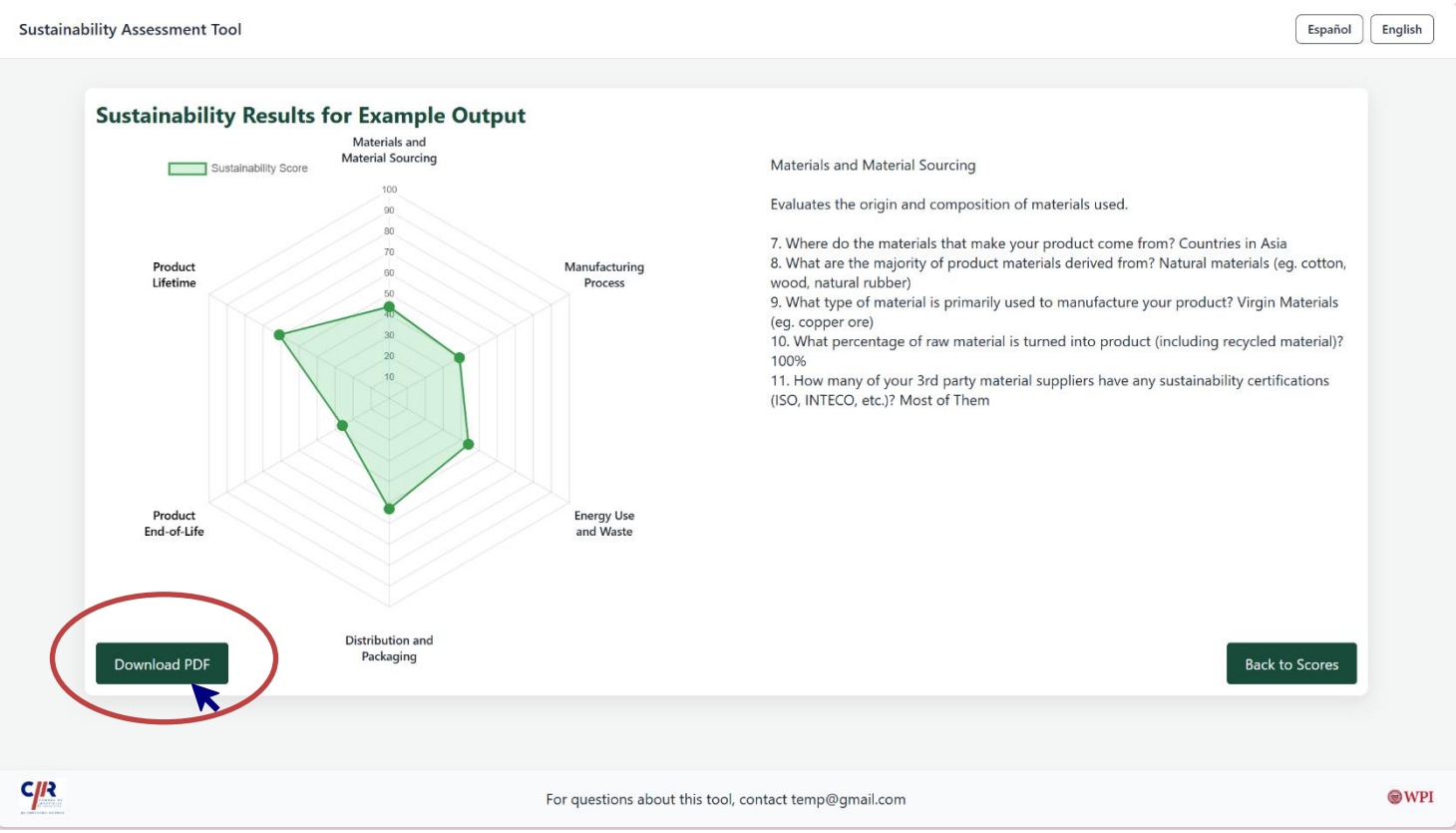
A larger, more evenly shaped polygon suggests balanced sustainability performance across categories, whereas an uneven shape highlights strengths and weaknesses in specific areas. The radar chart should be used as a comparative and diagnostic tool—it does not simply show a single “good” or “bad” score but rather illustrates how performance is distributed across different aspects of the product lifecycle.

How to Download the Results:

The results page can be easily saved and shared by selecting the “Download PDF” button. Clicking this button will generate a downloadable PDF document that includes all information displayed on the results

page, including the product title, radar chart, category scores, and detailed breakdowns. This feature allows users to securely store their results for future reference, internal documentation, reporting purposes, or comparison with future assessments.

If you would like to download the results in a different language, first switch the platform to your desired language using the language selection option (Page 3). Once the page has refreshed in the selected language, click “Download PDF” again to generate a new version of the report in that language. The PDF will reflect the currently selected language at the time of download.



Example PDF Output:

Each generated PDF file will be automatically saved using the following naming format:

PRODUCT_NAME_Sustainability_Assessment_Report_YEAR-DATE-MONTH.pdf

This standardized naming convention makes it easy to organize, archive, and retrieve reports at a later time. Including the product name and date in the file name helps prevent confusion when multiple assessments are conducted or when reports are shared among team members. Each PDF file is approximately 700 KB in size, making it small enough for convenient storage and email distribution while still maintaining clear graphics and formatting.

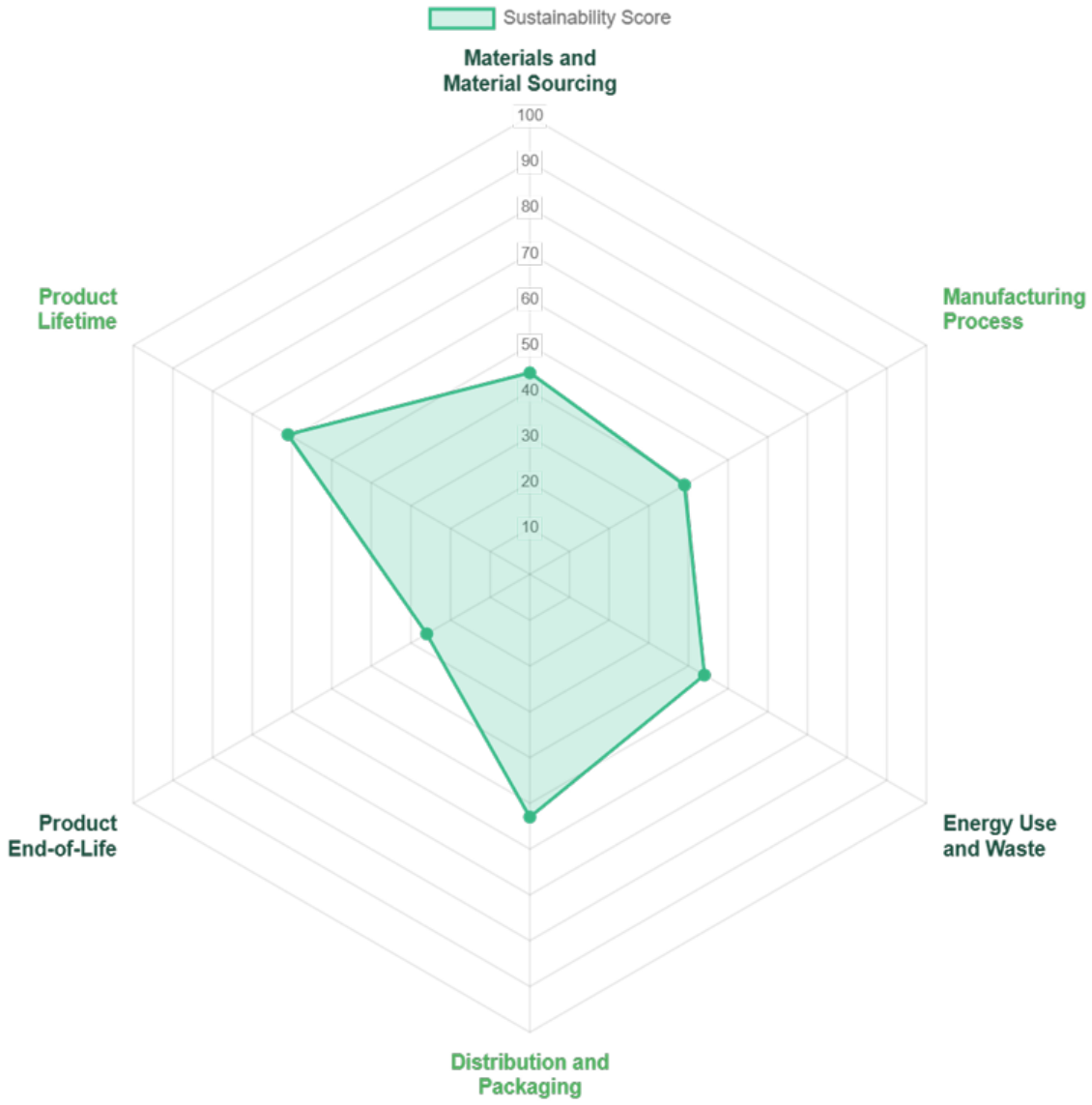
The first page of the PDF contains a summary overview of the assessment results. This includes the radar chart displaying performance across all six sustainability categories, as well as a clear listing of your product’s numerical score for each category. This page serves as a high-level snapshot of overall sustainability performance and is intended for quick review or executive summary purposes.

This can be seen on the next page:

Sustainability Assessment Report

Example Output

02/03/2026



Category Scores

Materials and Material Sourcing: 44
Energy Use and Waste: 44
Product End-of-Life: 26
Manufacturing Process: 39
Distribution and Packaging: 53
Product Lifetime: 61

The remaining pages of the PDF provide a complete record of the assessment. This section includes every question presented throughout the evaluation, along with the specific answers you selected, including demographic questions.

Example Page:

Sustainability Assessment Report

Detailed Responses

1. Which product would you like to analyze?
   Example Output

2. What is the size of your company?
   Small (6-30 employees)

3. What province is your company located (select multiple if your company has multiple locations)?
   Guanacaste

4. What sector is your product in (please reference CIIU Version 4 numbering chart)
   Example Output Product

5. Does your company have a sustainability strategy that is integrated into your overall business strategy?
   Yes

6. List all certifications or recognitions your company has. For example ISO 9001, ISO 14001, ISO 14064/INTE B5, or PBAE (Bandera Azul Ecológica). Write none if your company has none, you do not know, or if the question is not applicable.
   Example Output Certifications

7. Where do the materials that make your product come from?
   Countries in Asia

8. What are the majority of product materials derived from?
   Natural materials (eg. cotton, wood, natural rubber)

9. What type of material is primarily used to manufacture your product?
   Virgin Materials (eg. copper ore)

10. What percentage of raw material is turned into product (including recycled material)?
    100%

11. How many of your 3rd party material suppliers have any sustainability certifications (ISO, INTECO, etc.)?
    Most of Them

12. How is your equipment cleaned?
    With water

13. How often must your equipment be cleaned?
    Several times per week

14. Does your product require special safety procedures during manufacturing?
    No

15. Do you use natural or renewable resources during manufacturing?
    Rainwater

16. Does your company do any of the following related to water?
    Track water usage per unit of product, Actively work to lower waste water during manufacturing

INDEX INFORMATION

Categories and Their Descriptions

1. **Materials and Material Sourcing**
   - Short Description: Evaluates the origin and composition of materials used.
   - Long Description: This category assesses the environmental and ethical impacts associated with the materials used to create a product. It examines how materials are sourced, whether they are mined, farmed, or manufactured, and evaluates the transparency of the supply chain. Consideration is given to whether suppliers follow sustainable and ethical practices, hold certifications, and provide clear information about material origins.

     It also evaluates the renewability and composition of materials, such as whether they are petroleum-based, bio-based, recycled, or virgin resources. Additional factors include material scarcity, toxicity, and hazard potential, as well as whether materials are sourced locally or internationally. Together, these criteria determine how responsibly and sustainably a product's material foundation is established.

2. **Manufacturing Process**
   - Short Description: Evaluates efficiency and impact of the manufacturing process.
   - Long Description: This category examines how a product is manufactured and the environmental and operational impacts associated with production. It evaluates the complexity of the required equipment, the energy intensity of the process, and whether manufacturing is centralized or distributed across multiple facilities. Consideration is given to production scales such as small batch versus mass production, and how factory location influences transportation needs and overall environmental footprint.

     The category also assesses process efficiency, defect rates, and how manufacturing waste or defective products are managed. Efficient systems that minimize material waste, reuse scrap, and reduce energy consumption score more favorably. Overall, this category determines how responsibly and efficiently a product is brought from raw materials to finished form.

3. **Energy Use and Waste**
   - Short Description: Evaluates waste, water, and energy management.
   - Long Description: This category assesses waste generation, emissions, and energy use associated specifically with manufacturing and processing. It examines material yield (the ratio of input materials to finished product) to determine overall efficiency, as well as the volume and type of solid waste, liquid waste, and wastewater produced. Consideration is given to whether waste materials can be repurposed, recycled, or reused, or if they must be disposed of in landfills or through incineration.

     The category also evaluates air emissions, greenhouse gas output, and the presence of hazardous or toxic byproducts that may pose environmental or human health risks. Additionally, it considers how facilities treat waste, mitigate pollution, and comply with environmental regulations. Strong performance in this area reflects efficient energy use,

minimized waste streams, and responsible management of environmental impacts during production.

4. **Distribution and Packaging**
   - Short Description: Evaluates transportation packaging and logistics.
   - Long Description: This category examines the environmental impact of packaging and the systems used to transport products from manufacturing facilities to vendors and consumers. It evaluates the origin and type of packaging materials, including whether they are made from recycled or virgin resources, and whether they are recyclable, reusable, or compostable. Packaging volume and design efficiency are also considered, as excessive or poorly optimized packaging can increase material waste and transportation emissions.

     In addition, this category assesses product distribution distance, transportation methods (such as truck, ship, rail, or air), and whether vehicles are electric, hybrid, or fuel powered. The use of third-party logistics versus internal distribution systems, along with warehousing and storage impacts, are also taken into account. Overall, this category determines how efficiently and sustainably a product moves through the supply chain before reaching the end user.

5. **Product Lifetime**
   - Short Description: Evaluates durability and expected lifetime.
   - Long Description: This category assesses the sustainability of a product during its use phase, focusing on how long it remains functional and how easily it can be maintained or extended. It evaluates the expected product lifespan, durability under normal operating conditions, and common failure modes. Products designed for long-term use with robust materials and thoughtful engineering score higher than those intended for short-term or disposable use.

     The category also considers repairability, availability of replacement parts, and upgradability, including whether individual components can be replaced to extend overall lifespan. Safety during use is evaluated, along with whether customers can safely perform maintenance or if specialized tools and expertise are required. Manufacturer support, warranties, and access to repair documentation further reflect a company's commitment to extending product life and reducing premature disposal.

6. **Product End-of-Life**
   - Short Description: Evaluates product recyclability and end-of-life planning.
   - Long Description: This category assesses the environmental impact of a product once it reaches the end of its usable life, whether through normal wear or premature failure. It evaluates the product's recyclability, biodegradability, and ease of disassembly, considering whether materials can be separated efficiently for recycling or reuse. Products designed with clear material labeling and modular construction typically support more sustainable disposal pathways.

     The category also examines whether any components must be sent to landfills, require special handling due to hazardous materials, or pose environmental risks at disposal.

Additionally, it considers the availability of manufacturer take-back or recycling programs and how responsibly premature failures are managed. Strong performance in this area reflects thoughtful end-of-life planning that minimizes waste and reduces environmental harm.

List of Questions (Points + Explanations)

Question Legend:

- ○ Multiple-Choice: This icon represents a multiple-choice question where only one of the options may be selected. The score associated with your answer will be the score you receive.
- ☐ Checkbox: This icon represents a checkbox question. Multiple answers may be selected for the same question. The scoring for these types of questions is explained in the Scoring Explanation section.

Text Box

This icon represents a text box entry. Anything entered will be saved and sent to CICR's database as the answer. No score is applied to these questions.

Slider: This icon represents a slider question. Each slider will have a minimum and a maximum that will each represent a score. An answer selected between the min and max will have a score proportional to the percentage of the distance from the minimum to the maximum that the slider is set at.

Demographic Questions (Not Scored):

1. What is the name of your product?

Text Box

Explanation: This question is included mainly to aid in searching and organizing for both CICR and individual companies. The PDF is saved with a name that includes the product name, allowing companies to easily organize result PDFs by product. It also allows CICR to search for specific products if a company requests consultation about a particular product. This makes it easier to track that product's performance over time and review the submitted responses associated with it.

2. What is the size of your company?
   - ○ Micro (1-5 employees)
   - ○ Small (6-30 employees)

- Medium (31-120 employees)
- Large (121+ employees)

Explanation: This question is included to allow CICR to track trends across companies of different sizes. By analyzing responses based on company size, CICR can identify patterns, strengths, and areas where certain size companies may need additional support. This helps them better tailor resources, guidance, and consultation efforts to the specific needs of different types of companies.

3. What province is your company located (select multiple if your company has multiple locations)?

- ☐ Alajuela
- ☐ Cartago
- ☐ Guanacaste
- ☐ Heredia
- ☐ Limón
- ☐ Puntarenas
- ☐ San José

Explanation: This question is included to allow CICR to track trends across companies from different provinces. By analyzing responses based on provinces, CICR can identify patterns, strengths, and areas where certain genres of companies may need additional support. This helps them better tailor resources, guidance, and consultation efforts to the specific needs of different types of companies.

4. What sector is your product in (please reference CIIU numbering chart: https://www.hacienda.go.cr/docs/CorrespondenciaATVCIIU3-TRIBUCRCIIU4-Completo.pdf**)**

Text Box

Explanation: This question is included mainly to aid in searching and organizing for CICR. It aids CICR search to find a specific product if a company requests consultation about a particular product and the product name is not found. This makes it easier to track that product's performance over time and review the submitted responses associated with it. It also allows CICR to track trends from companies in the same or similar sectors. By analyzing responses based on sectors, CICR can identify patterns, strengths, and areas where certain genres companies may need additional support. This helps them better tailor resources, guidance, and consultation efforts to the specific needs of different types of companies.

5. Does your company have a sustainability strategy that is integrated into your overall business strategy?

   - Yes
   - No

   Explanation: This question is included to understand how many companies have formally integrated sustainability into their overall business strategy. It helps CICR gauge how common structured sustainability planning is among participating organizations.

6. List all certifications or recognitions your company has. For example, ISO 9001, ISO 14001, ISO 14064/INTE B5, or PBAE (Bandera Azul Ecológica). Write none if your company has none, you do not know, or if the question is not applicable.

Text Box

   Explanation: This question allows CICR to track trends in certifications and recognitions across companies. It helps identify commonly held certifications, emerging standards, and new or growing sustainability credentials within the industry.

Scored Questions:

Materials and Material Sourcing Questions:

7. Where do the materials that make your product come from? (Averaged Checkbox Type)

   - ☐ Costa Rica (10pts)
   - ☐ Countries Within North, Central, or South America (5pts)
   - ☐ Countries in Asia (0pts)
   - ☐ Other countries (0pts)
   - ☐ Not applicable / Not sure (0/0pts)

   Explanation: This question evaluates the geographic sourcing of materials. It is designed to encourage more localized or regional sourcing, which can reduce transportation emissions, and improve supply chain transparency. Higher scores are awarded to materials sourced closer to the point of production (Costa Rica), while materials sourced from more distant regions receive fewer points due to increased transportation impacts and potential supply chain complexity.

8. What are your product's materials derived from? (Averaged Checkbox Type)

   - ☐ Recycled/reclaimed materials (eg. recycled aluminum) (10pts)
   - ☐ Renewable/bio-based materials (eg. wood) (10pts)
   - ☐ Virgin mineral or metal materials (eg. newly mined iron) (5pts)

- ☐ Virgin fossil based synthetic materials (eg. plastic, nylon) (0pts)
- ☐ Not applicable/Not sure (0/0pts)

Explanation: This question assesses the origin and sustainability of the materials used in the product. Higher scores are given to recycled, reclaimed, and renewable materials, as these typically reduce resource extraction and environmental impact. Virgin mineral or metal materials receive moderate points due to their higher extraction impact, while virgin fossil-based synthetic materials receive lower scores due to their reliance on non-renewable resources and associated environmental concerns.

9. Does your product rely on scarce materials?

- ○ Yes, all of the materials are scarce (0pts)
- ○ Yes, most of the materials are scarce (2pts)
- ○ Yes, but only some of the materials are scarce (4pts)
- ○ Yes, but almost none of the materials are scarce (6pts)
- ○ No, all materials are abundant (10pts)
- ○ Not applicable/Not sure (0/0pts)

Explanation: This question evaluates the degree to which a product depends on scarce or limited resources. Materials that are rare, difficult to source, or geopolitically sensitive can pose environmental, economic, and supply chain risks. Products that rely heavily on scarce materials receive lower scores due to these long-term sustainability concerns.

10. What percentage of raw material is eventually turned into a product (eg. If buy 100kg of material, does all 100 kg eventually turn into a product)?

- ○ 100% (10pts)
- ○ 90-99% (9pts)
- ○ 75-89% (7pts)
- ○ 50-75% (5pts)
- ○ Less than 50% (0pts)
- ○ Not applicable / Not sure (0/0pts)

Explanation: This question measures material efficiency during production, often referred to as material yield. It evaluates how much of the input material becomes part of the final product rather than being lost as scrap or waste. Higher percentages indicate more efficient use of resources and reduced waste generation, resulting in higher scores.

11. How many of your material suppliers have any sustainability certifications (ISO/INTECO, etc.)?

- ○ All of Them (10 pts)
- ○ Most of Them (8pts)
- ○ Half of Them (5pts)
- ○ Some of Them (3pts)
- ○ None of Them (0pts)
- ○ Not applicable / Not sure (0/0pts)

Explanation: This question assesses the sustainability practices within a company's supply chain. Suppliers with recognized certifications (such as ISO or INTECO standards) demonstrate adherence to environmental management systems or sustainability frameworks. A higher percentage of certified suppliers indicates stronger supply chain accountability and environmental oversight, resulting in a higher score.

Manufacturing Process Questions:

12. What percentage of your manufactured products are defective?

Min: 0%, Max: 100%, (0% = 10pts, 100% = 0 pts)

Explanation: This question measures production quality and process efficiency. A lower defect rate indicates more precise manufacturing, better quality control, and reduced material waste. Because defective products often require rework or disposal, higher defect percentages result in lower scores. A defect rate of 0% represents optimal efficiency and earns the maximum score.

13. How is your equipment cleaned? (Averaged Checkbox Type)

- ☐ With non-consumable materials (ex: a brush) (10pts)
- ☐ With water (7pts)
- ☐ Using environmentally safe chemicals (6pts)
- ☐ Using other types of chemicals (3pts)
- ☐ A more intensive method(0/0pts)
- ☐ Not applicable/ Not sure (0pts)

Explanation: This question evaluates the environmental impact of equipment cleaning practices. Cleaning methods that use fewer consumables and less hazardous substances score higher, as they typically generate less waste and reduce chemical exposure risks. More resource-intensive or

chemically intensive methods receive lower scores due to their greater environmental and health impacts.

14. How often must your equipment be cleaned?
    - Multiple times per day (0pts)
    - Once a day (1pts)
    - Several times per week (2pts)
    - Once per week (3pts)
    - Once per month (4pts)
    - Every 3 or 4 months (6pts)
    - Once per year (8pts)
    - Once every 2+ years (10pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question assesses the resource intensity associated with equipment maintenance. More frequent cleaning often indicates higher water use, chemical use, labor, and potential waste generation. Less frequent cleaning requirements generally reflect more efficient processes or equipment designs, resulting in higher scores.

15. Does your product require any of the following procedures for employees during manufacturing? (Averaged Checkbox Type)

    - ☐ Extra safety protocol for equipment use (8pts)
    - ☐ Decontamination of materials or employees (5pts)
    - ☐ Minor protective equipment (eg. Masks or gloves) (5pts)
    - ☐ Major protective equipment (eg. hazard suits) (0pts)
    - ☐ None (10pts)
    - ☐ Not applicable/Not sure (0/0pts)

    Explanation: This question evaluates the level of hazard associated with the manufacturing process. If significant protective measures or decontamination procedures are required, it may indicate higher environmental risks.

16. Do you use renewable resources to help product manufacturing? (Capped Checkbox Type)

- ☐ Sunlight (2pts)
- ☐ Rainwater (2pts)
- ☐ Hydropower (2pts)
- ☐ Windpower (2pts)
- ☐ Natural ventilation (2pts)
- ☐ Other (2pts)
- ☐ None (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question assesses the integration of renewable or low-impact resources into the manufacturing process. Each renewable resource used contributes positively toward the total score. Encouraging the use of multiple renewable resources supports more sustainable manufacturing practices and reduces reliance on non-renewable energy or materials.

Energy Use and Waste Questions:

17. Does your company do any of the following related to water? (Capped Checkbox Type)

- ☐ Conduct regular water audits (1–2 years) (2pts)
- ☐ Track water usage per unit of product (2pts)
- ☐ Treat wastewater before disposal (onsite or through a 3rd party company) (2pts)
- ☐ Actively work to lower wastewater during manufacturing (2pts)
- ☐ Measure water quality regularly (2pts)
- ☐ Recycle wastewater (2pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates how actively the company monitors and manages water usage and wastewater. Practices such as audits, tracking, treatment, and recycling indicate proactive management of water resources and reduced environmental impact. Higher engagement in these practices reflects better sustainability performance.

18. Does your company do any of the following related to energy? (Capped Checkbox Type)

- ☐ Conduct regular energy audits (3–5 years) (2pts)
- ☐ Track energy usage per unit of product (2pts)
- ☐ Actively work to lower energy usage during manufacturing (2pts)

- ☐ Generate renewable energy for self-consumption (2pts)
- ☐ Reuse energy (eg. use rechargeable batteries, use heat emissions as energy) (2pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question assesses energy management practices. Tracking energy use, conducting audits, and implementing renewable or reused energy solutions indicate efficient and sustainable energy practices. Companies with more comprehensive energy management are recognized with higher scores.

19. Does your company do any of the following related to materials? (Capped Checkbox Type)

- ☐ Track the amount of material waste per product (2pts)
- ☐ Collect and recycle waste material for every or most stages of product creation (2pts)
- ☐ Recycle or repurpose all or most defective products (2pts)
- ☐ Recycle manufacturing waste (by yourself or with another company) (2pts)
- ☐ Design products or manufacturing processes to minimize material waste (2pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates how effectively a company monitors, reduces, and recycles material waste. Tracking and repurposing waste, as well as designing processes to minimize waste, demonstrates strong sustainability practices.

20. Over the past 5 years has energy or electricity consumption to manufacture this product gone down?

- ○ Yes, fuel consumption was reduced (10pts)
- ○ Yes, energy consumption was reduced (10pts)
- ○ No (0pts)
- ○ Not applicable / Not sure (0/0pts)

Explanation: This question assesses whether the company has improved energy efficiency over time. A reduction in energy consumption reflects efforts to optimize production processes, reduce emissions, and decrease operational costs, all of which contribute positively to sustainability performance.

21. Over the past 5 years has water consumption to manufacture this product gone down?

- ○ Yes (10pst)
- ○ No (0pts)

- Not applicable / Not sure (0/0pts)

Explanation: This question measures improvements in water efficiency. Reduced water use indicates better resource management and less environmental impact, contributing to higher sustainability performance.

22. If you generate your own energy, is it from renewable sources (eg. wind turbines)?

- Yes (10pts)
- No (0pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the use of renewable resources for self-generated energy. Utilizing renewable sources reduces reliance on non-renewable energy and decreases environmental impacts, contributing positively to the company's sustainability score.

23. If you collect your own water, is it from renewable sources (eg. rainwater)?

- Yes (10pts)
- No (0pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the use of renewable resources for self-generated water. Utilizing renewable sources reduces reliance on non-renewable energy and decreases environmental impacts, contributing positively to the company's sustainability score.

24. What air pollutants/greenhouse gases are released as part of your manufacturing process? (Averaged Checkbox Type)

- ☐ Not measured (0pts)
- ☐ None (10pts)
- ☐ Carbon Dioxide (5pts)
- ☐ Nitrous Oxide (5pts)
- ☐ Methane (5pts)
- ☐ Fluorine containing gases (0pts)
- ☐ Other high global warming potential refrigerants or chemical fumes (0pts)
- ☐ Other (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question assesses the type and extent of emissions produced during manufacturing. Companies that measure and minimize emissions score higher, while those that release significant greenhouse gases or do not measure their emissions receive lower scores.

25. If you release any greenhouse gases/pollutants, do you have a process for limiting it?

    - Yes (10pts)
    - No (0pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question evaluates whether the company actively manages and mitigates emissions. Implementing processes to limit emissions demonstrates responsible environmental management and contributes positively to sustainability performance. If no greenhouse gas emissions are created, users should select "Not applicable."

26. Is there hazardous waste created during the manufacturing process?

    - Yes (0pts)
    - No (10pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question assesses whether manufacturing generates hazardous waste. Avoiding hazardous waste production reduces environmental and earns higher scores.

27. If hazardous waste is created, do you properly dispose of it (eg. sent to a disposal factory)?

    - Yes (10pts)
    - No (0pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question evaluates whether any hazardous waste generated is managed responsibly. Proper disposal of hazardous materials prevents environmental contamination and health risks, reflecting strong operational sustainability practices. If no hazardous waste is created, users should select "Not applicable."

Distribution and Packaging Questions:

28. Does your product require multiple factory sites to create?

    - Yes (0pts)
    - No (10pts)

- ○ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the complexity and efficiency of the production and distribution network. Products that require multiple factory sites typically involve additional transportation, coordination, and resource use, which increases environmental impacts. Single-site production is considered more streamlined and earns higher scores.

29. What kind of vehicles does your company use to transport the product? (Averaged Checkbox Type)

- ☐ Electric vehicles (10pts)
- ☐ Hybrid vehicles (electric/gasoline/diesel) (8pts)
- ☐ Hybrid vehicles (lpg/gasoline/diesel) (6pts)
- ☐ Gas vehicles (2pts)
- ☐ Diesel vehicles (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question assesses the environmental impact of product transportation. Using cleaner vehicle technologies, such as electric or hybrid vehicles, reduces greenhouse gas emissions and other pollutants, resulting in higher scores. Traditional fossil fuel vehicles contribute more to environmental impacts and a lower score. If your product does not require transportation or your company does not transport the product (eg. clients come pick it up themselves), users should select "Not applicable."

30. What percentage of your products produced is exported outside of Costa Rica?

Min: 0%, Max 100% (0=10pts, 100=0pts)

Explanation: This question evaluates the carbon and environmental footprint associated with product distribution. Products that are primarily exported require longer transportation distances, leading to increased energy use and emissions. A lower export percentage reflects shorter supply chains and lower environmental impact, earning higher scores.

31. How long does it take for the majority of products to reach clients within Costa Rica?

- ○ 1-4 hours (10pts)
- ○ One day (8pts)
- ○ Half a week (5pts)
- ○ One week (3pts)

- Longer than a week (0pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question assesses logistical efficiency and the environmental impact of domestic transportation. Faster delivery times generally indicate shorter supply chains, lower energy use, and reduced emissions, contributing to higher scores. Longer delivery times may reflect inefficiencies or additional transportation requirements, resulting in lower scores.

32. What is the primary origin of the project's packaging? (Averaged Checkbox Type)

- ☐ Created in the same factory as the product (10pts)
- ☐ Costa Rica (10pts)
- ☐ Countries Within North, Central, and South America (5pts)
- ☐ Countries in Asia (0pts)
- ☐ Other countries (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the sourcing of packaging materials. Packaging sourced locally or produced on-site reduces transportation-related emissions and supply chain complexity, earning higher scores. Packaging sourced from distant countries typically requires longer transport routes, increasing environmental impacts and resulting in lower scores.

33. Is your product's packaging all or partially: (Averaged Checkbox Type)

- ☐ Recyclable (10pts)
- ☐ Biodegradable or compostable (10pts)
- ☐ Reusable/returnable (10pts)
- ☐ Must be disposed of in the garbage (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the sustainability of the product’s packaging. Packaging that can be recycled, composted, or reused reduces waste and environmental impact, earning higher scores. Packaging that must be disposed of as garbage has a higher environmental footprint and results in lower scores.

Product Lifetime Questions:

34. Are your products stored in a way that can cause environmental hazards (ex: outside uncovered/unsealed)?

- Yes (0pts)
- No (10pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question evaluates potential environmental risks associated with product storage. Improper storage that exposes products to the environment can lead to contamination, spills, or material degradation. Proper storage practices reduce these risks and support safer, more sustainable handling of products.

35. Does your product release any hazardous material during its lifetime?
    - Yes (0pts)
    - No (10pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question assesses whether the product poses environmental or health hazards during its use. Products that do not release hazardous materials are considered safer and more sustainable, earning higher scores. Products that release hazardous substances receive lower scores due to potential risks to humans and the environment.

36. Can your product be repaired or upgraded during its lifetime? Select not applicable if your product is a consumable.
    - Yes (10pts)
    - No (0pts)
    - Not applicable / Not sure (0/0pts)

    Explanation: This question measures durability and adaptability. Products that can be repaired or upgraded have a longer useful life, reducing waste and resource consumption. If the product is a consumable product, users should select “Not applicable.”

37. How many years is your product designed to last?
    - 0–11 months (2pts)
    - 1–5 years (5pts)
    - 6–10 years (8pts)
    - 11+ years (10pts)
    - Not applicable / Not sure (0/0pt)

    Explanation: This question evaluates the expected product lifespan. Longer-lasting products generally reduce the frequency of replacement, lower material consumption, and minimize waste,

contributing to higher sustainability performance. If the product is a consumable product, users should select “Not applicable.”

38. During its lifetime, which of the following ongoing inputs or maintenance requirements does your product need? (Averaged Checkbox Type)

    - ☐ Constant energy input (0pt)
    - ☐ Periodic energy input (chargeable) (5pt)
    - ☐ Regular cleaning or maintenance materials (eg. Lubricants, detergents) (3pt)
    - ☐ Temperature control (refrigeration) (2pts)
    - ☐ Replacement consumables (eg. Filters, batteries, cartridges) (0pts)
    - ☐ Other methods that require extra materials (2pt)
    - ☐ None (10pts)
    - ☐ Not applicable/ Not sure (0/0pts)

    Explanation: This question assesses the ongoing resource demands of a product during its use phase. Products that require frequent energy input, consumables, or maintenance materials typically have a higher environmental impact. Products that operate independently with minimal input are more sustainable and score higher.

Product End-Of-Life Questions:

39. Does your product need to be disassembled to dispose of properly?

    - ○ No (10pts)
    - ○ Yes, and can be done easily (5pts)
    - ○ Yes, but cannot be done easily (requires special equipment) (0pts)
    - ○ Not applicable / Not sure (0/0pts)

    Explanation: This question evaluates how easily a product can be prepared for proper disposal or recycling. Products that do not require disassembly are simpler for consumers to manage and are considered more sustainable. Products requiring complex or specialized disassembly create barriers to proper disposal and score lower.

40. When your product is disposed of, it is: (Averaged Checkbox Type)

    - ☐ Fully Recyclable (10pts)
    - ☐ Fully Compostable (10pts)
    - ☐ Fully Biodegradable (10pts)

- ☐ Fully Co-processable (2pts)
- ☐ Partially Recyclable (5pts)
- ☐ Partially Compostable (5pts)
- ☐ Partially Biodegradable (5pts)
- ☐ Partially Co-processable (1pts)
- ☐ Requires other special treatment (1pts)
- ☐ Must go to the landfill (0pts)
- ☐ Not applicable / Not sure (0/0pts)

Explanation: This question assesses the end-of-life sustainability of the product. Products that are fully recyclable, compostable, or biodegradable minimize environmental impact and waste. Partial or specialized disposal methods reduce sustainability performance, and products destined for landfill score lowest.

41. If your product is not biodegradable, compostable, or recyclable on its own do you recollect them from customers to dispose of properly?

- ○ Yes (10pts)
- ○ No, contact information is given for another company to do it (5pts)
- ○ No (0pts)
- ○ Not applicable / Not sure (0/0pts)

Explanation: This question evaluates responsibility for proper end-of-life management. Actively recollecting products ensures safe disposal or recycling and reflects strong environmental stewardship. Providing contact information for disposal is positive but less proactive, while no program in place scores lowest. If a product cannot be biodegradable, compostable, or recyclable at all, users should select “Not applicable.”

42. Does your product release any hazardous material or emissions if it fails?

- ○ Yes (0pts)
- ○ No (10pts)
- ○ Not applicable / Not sure (0/0pts)

Explanation: This question assesses potential environmental or health risks if the product fails. Products that do not release hazardous materials or emissions are safer and more sustainable, earning higher scores.

43. How many times can your product be reused or recycled? If your product is a consumable, select not applicable.

- 0 (0pts)
- 1-5 (5pts)
- 5-30 (7pts)
- Infinitely (10pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question evaluates the durability and circularity of the product. Products that can be reused or recycled multiple times reduce waste and resource demand, contributing to higher sustainability performance. If the product is a consumable product, users should select "Not applicable."

44. If your product is a consumable, do you minimize client waste (eg. offer bulk sizing options to reduce per-unit disposal)?

- Yes (10pts)
- No (0pts)
- Not applicable / Not sure (0/0pts)

Explanation: This question assesses efforts to reduce consumer-generated waste from consumable products. Minimizing per-unit disposal through bulk sizing or other strategies decreases environmental impact and earns higher scores. If the product is not a consumable product, users should select "Not applicable."

Scoring Explanation

This section explains how the scoring system is structured and calculated. Each question in the assessment is worth a maximum of 10 points. Questions are organized into six sustainability categories: Materials and Material Sourcing, Manufacturing Process, Energy Use and Waste, Distribution and Packaging, Product Lifetime, and Product End-of-Life.

Because each category contains a different number of questions, raw point totals are normalized to a 0–100 scale. This ensures that categories with more questions do not automatically carry more weight than those with fewer questions. A category score therefore represents the percentage of total possible points earned within that category, allowing for fair comparison across all six areas of evaluation.

Scoring Important Note:

Each question includes a "Not Applicable / Not Sure" option. When this option is selected, the question is scored as 0/0, meaning it does not contribute to either the earned points or the total possible points for that category. In other words, the question is excluded from the final score calculation.

For example, if a category originally contains five questions (50 total possible points) and one question is marked "Not Applicable / Not Sure," the category will instead be calculated as having four active questions (40 total possible points). This approach ensures that respondents are not penalized for questions that genuinely do not apply to their product or for situations where sufficient information is unavailable. The goal is to maintain fairness and accuracy in the evaluation process while preserving the integrity of the scoring system.

Scoring for Each Question Type:

Text:

The text box questions only occur in the demographic section of the tool. None of these questions are scored. Essentially having a score of 0/0.

Multiple-choice:

Multiple-choice questions in the demographic section are not scored. All other multiple-choice questions throughout the assessment are scored in the same manner as standard evaluation questions. Each question is worth up to 10 points, and the selected response determines how many points are awarded based on the scoring criteria assigned to each option. These points are then incorporated into the appropriate category total and included in the overall normalization process.

Checkbox:

Checkbox questions use a slightly different scoring method than other question types. Because multiple options can be selected, two scoring approaches are used: "Capped" and "Averaged." The scoring type is determined by the design of the specific question.

Capped:

For capped checkbox questions, the point values of all selected options are added together. However, the total score for that question cannot exceed 10 points. If the sum of selected options exceeds 10, the score is capped at 10. This approach rewards selecting multiple applicable positive options while preventing the total from exceeding the maximum allowed score for a single question.

Averaged:

For averaged checkbox questions, the point values of all selected options are added together and then divided by the total number of options selected. This produces an average score for that question, up to a maximum of 10 points. This method is used when the intention is to evaluate the overall quality of selected practices rather than simply rewarding the quantity of selections.

For example, if one option one selected is worth 10 points but another option selected worth 5 points, one would receive a score of 7.5 points.

Slider:

Slider questions allow users to select a value along a defined numerical range. Each slider question is worth up to 10 points, and the selected value is scaled appropriately within the scoring system. The

score is proportional to the percentage of the distance from the minimum to the maximum that the slider is set at.

In some cases, 100% represents positive performance (for example, 100% recycled content), meaning higher percentages correspond to higher point values (100% = 10 points). In other cases, 100% represents negative performance (for example, 100% landfill waste), meaning higher percentages correspond to lower point values (100% = 0 points).

Whether a slider is scored normally or inversely depends on how the minimum and maximum point values are defined in the manual. If the minimum point value is greater than the maximum point value (min > max), the slider is automatically treated as inverted. This ensures that scoring remains consistent and aligned with the sustainability intent of each question, regardless of whether a higher percentage represents better or worse performance.

Final Score Calculation:

The final score is calculated on a category-by-category basis. For each of the six categories (Materials and Material Sourcing, Manufacturing Process, Energy Use and Waste, Distribution and Packaging, Product Lifetime, and Product End-of-Life) the total points earned are summed and compared to the total possible points within that category.

Because categories may contain different numbers of questions, each category score is normalized to a 0–100 scale. This is done by dividing the total points earned in a category by the total possible points for that category (excluding any questions marked “Not Applicable / Not Sure”) and then multiplying by 100. The math goes:

Raw Total Points Scored/Raw Maximum Points * 100 = Final Category Score

This ensures that all categories are weighted equally in presentation, regardless of how many questions they contain. The final displayed results therefore represent percentage-based performance in each sustainability category, allowing for fair comparison and clear interpretation across all areas of the assessment.

OTHER INFO

About Us (With Contacts)

Worcester Polytechnic Institute (WPI) is a technological university founder in 1865 in Worcester, Massachusetts, United States of America. It focusses on applying theory learned in the classroom to solving real-world problems through projects.

One of these required projects is the **Interactive Qualifying Project (IQP)**. It is an award winning, global project where students, typically 3rd year students, solve a scientific, technological, and/or societal problem through a public or private sponsor. There are 50 WPI project centers across the globe.

This project was an IQP sponsored by CICR, specifically guided by Sustainability, Environment, and Social Responsibility Advisor at CICR, Akira Hidalgo-Segura. The advisors for the project were Dr. Lina Muñoz-Márquez and Dr. Alejandro Manga.

The IQP team that created this tool was comprised of 4 third year students:

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FAQs (Frequently Asked Questions)

1. **How do I begin using the Product Sustainability Assessment Tool?**
   - You can start at the Landing Page, which explains the purpose of the tool, confidentiality terms, and how it works. Moving past this page counts as an electronic signature agreeing to the terms.

   - *Please consult the section titles "How to Fill One Out" starting at page 3 for full instructions.*

2. **How long will it take to complete the assessment?**
   - The assessment currently has approximately 44 questions and typically takes 20–30 minutes to finish.

   - *Please consult the section titles "How to Fill One Out" starting at page 3.*

3. **Can I fill out the tool in a language other than English?**
   - Yes. You can switch between English and Spanish at any time, and the platform will live-update previous answers.

   - *Please consult page 3.*

4. **Are all questions required to calculate my product's score?**
   - No. You can select "Not applicable / Not sure" for questions that do not apply. These are scored as 0/0 and are excluded from the final score calculation.

   - *Please consult page 30.*

5. **How do I answer the different question types?**
   - Textbox: Type your answer and click next. These are not scored.
   - Multiple-choice: Select one answer; scores vary depending on the choice.
   - Checkbox: Select one or more answers. Scores are calculated using either "capped" or "averaged" methods.
   - Slider: Move the slider along a range; the score is proportional to the position.

   - *Please consult 4 and 31.*

6. **How is my product's sustainability score calculated?**
   - Each question is worth up to 10 points. Scores are totaled per category and normalized to a 0–100 scale, taking into account any "Not applicable / Not sure" answers.

   - *Please consult page 30.*

7. **What do the radar chart and score panel represent?**

- The radar chart visualizes your product's performance across six sustainability categories. The score panel displays exact numerical scores. Both help identify strengths and areas for improvement.

- *Please consult page 6.*

8. Can I download my results?

- Yes. You can download a PDF of your assessment, including the radar chart, category scores, and detailed answers. If you switch the language, the PDF will reflect the selected language.

- *Please consult page 9.*

9. Who can access the data I provide?

- Only select CICR representatives can access the data for statistical reporting. Identifying information about your product or company will not be published. You control whether results are shared externally.

- *Please consult page 3.*

10. How do I know which questions belong to each sustainability category?

- Each question is associated with one of six categories: Materials and Material Sourcing, Manufacturing Process, Energy Use and Waste, Distribution and Packaging, Product Lifetime, and Product End-of-Life. You can view the breakdown on the results page.

- *Please consult page 15.*

11. What should I do if my product is a consumable?

- Some questions about reuse, recycling, or waste reduction may not apply. In such cases, select "Not applicable / Not sure" or answer regarding minimizing client waste.

- *Please consult the section "Scored Questions" starting on page 17.*

12. What happens if my product generates hazardous waste or emissions?

- There are specific questions to evaluate hazardous materials and emissions management. Proper disposal or mitigation increases your sustainability score.

- *Please consult the section "Scored Questions" starting on page 17.*

13. Who created this tool and who can I contact?

- The tool was developed by WPI students under the guidance of CICR. Contact emails for the team and CICR advisors are available in the "About Us" section.

- *Please consult page 34.*

14. Can I correct or change my responses after submission?

- Once submitted, responses are saved in CICR's database. You may need to contact CICR directly to correct any errors.

- *Please consult page 3 for confidentiality and data storage policies and page 34 for contact information.*